Compass Project Communication Protocol

Version 1.0.0

Benjamin Huh, Oliravan Eswaramoorthy, Jason Peng

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Contents

1 General Information

1.1 Abbreviations and Definitions

- **BLE** - Bluetooth Low Energy
- **LoRA** - Long Range radio communication protocol
- **RSSI** - Received Signal Strength Indicator
- **CAD** - Channel Activity Detection
- **TTL** - Time To Live
- **PSF** - Path So Far
- **LHGPS** - Last Heard GPS location
- **GPS** - Global Positioning System
- **BLE Edge** - A Bluetooth device that is not a Clusterhead; a leaf node in the BLE mesh
- **BLE Clusterhead** - A Bluetooth device that coordinates communication within a BLE cluster
- **LoRA Edge** - A LoRA device that is not a Clusterhead
- **LoRA Clusterhead** - A LoRA device that coordinates communication between LoRA clusters
- **Node/Device** - A single unit in the mesh network (terms used interchangeably)
- **Cluster** - A group of nodes that communicate directly with each other and share a Clusterhead
- **Mesh Network** - A network topology where nodes relay data for the network
- **Crowding Factor** - A metric indicating the density of active devices in a given area or channel
- **Critical Nodes** - Nodes that serve essential roles in the network, such as Clusterheads or nodes that bridge otherwise disconnected network segments

- **Affirmation** - A confirmation message sent by a Clusterhead to an Edge node acknowledging the Edge's presence in the cluster and providing path information

1.2 Brief Summary and Scope of the Protocol

This document describes the hybrid Bluetooth and LoRA communication protocol for the compass project. It was designed to be used in a densely packed environment with either little to no WiFi and Cellular Signal or overloaded telecommunications infrastructure.

It is designed to be functionally robust in environments with high interference while allowing devices to communicate time-sensitive information such as geographical location within 30 seconds. It is also designed to handle a potential load of up to 1,000 devices per mesh network, with individual Bluetooth clusters supporting up to 150 devices and LoRA channels supporting up to 32 devices each.

1.3 Conventions

Throughout this document, the following conventions are used:

- Message formats are described as comma-separated fields (e.g., "ID, TTL, PSF, LHGPS")
- Time measurements are in seconds unless otherwise specified
- "Cycle" refers to one complete iteration of a discovery or communication phase
- "Slot" refers to a time-division multiplexing window allocated for transmission or reception
- "Forwarding" means relaying a message received from one node to other nodes in the network
- "Flooding" refers to broadcasting a message to all reachable nodes
- "Managed flooding" means controlled message propagation using metrics like TTL, crowding factor, or GPS proximity
- Channel numbers (k) for LoRA start at 1 and increase based on network density
- Percentages and thresholds (e.g., "90%", "75%") represent target values that may be adjusted based on deployment conditions

2 Protocol Description

2.1 Bluetooth Mesh Specifications

The Bluetooth mesh operates using BLE (Bluetooth Low Energy) technology with the following specifications:

- **Operating Frequency**: 2.4 GHz ISM band
- **Maximum Cluster Size**: 150 devices per BLE cluster
- **Discovery Channel**: Single shared channel for all BLE discovery operations
- **Message Slots per Discovery Cycle**: 4 slots (1 for own message, 3 for forwarding)
- **Clusterhead Listening Time**: 75% of messaging phase
- **Clusterhead Messaging Time**: 25% of messaging phase
- **Network Topology**: Hierarchical mesh with BLE Edges connecting to BLE Clusterheads
- **Routing Strategy**: Path-based routing with managed flooding during discovery

2.2 LoRA Mesh Specifications

The LoRA mesh provides long-range communication with the following specifications:

- **Number of Channels**: Up to 56 channels (dynamically allocated)
- **Maximum Devices per Channel**: 32 devices
- **Channel Selection**: Dynamic based on crowding factor and RSSI
- **Initial Channel**: All devices start on channel 1
- **Network Topology**: Clustered mesh with LoRA Edges connecting to LoRA Clusterheads
- **Communication Range**: Extended range compared to Bluetooth (implementation-dependent)
- **Data Rate**: Lower than BLE, suitable for control messages and location data
- **Channel Migration**: Voting-based mechanism for load balancing across channels

2.3 Modes 1 and 2

The protocol operates in a pipelined fashion, alternating between two distinct modes that run simultaneously across BLE and LoRA radios. Each mode takes 1.5 seconds, and the modes flip-flop continuously:

- **Mode 1 - Clusterhead $\leftrightarrow$ Edge Communication (1.5s)**:
  - **BLE**: BLE Clusterheads and BLE Edges communicate bidirectionally within their clusters
  - **LoRA (simultaneous)**: LoRA Clusterheads communicate with other LoRA Clusterheads
  - Edge to CH and CH to Edge communication both occur within this mode
- **Mode 2 - Complementary Communication (1.5s)**:
  - **BLE**: BLE Clusterheads communicate with other BLE Clusterheads
  - **LoRA (simultaneous)**: LoRA Clusterheads and LoRA Edges communicate bidirectionally within their clusters
  - Edge to CH and CH to Edge communication both occur within this mode

The protocol continuously cycles: Mode 1 (1.5s) $\rightarrow$ Mode 2 (1.5s) $\rightarrow$ Mode 1 (1.5s) $\rightarrow$ Mode 2 (1.5s)... This pipelined operation ensures both BLE and LoRA radios are active simultaneously, with different message types on each protocol to maximize bandwidth utilization and minimize latency.

2.4 Submodes for Rediscovery

During normal Mode 1 and Mode 2 operation, the protocol includes two submodes that execute every 30 minutes. When the 30-minute mark is reached, the protocol waits until the appropriate mode cycle occurs before triggering the rediscovery submode:

- **Submode 1 - BLE Rediscovery**:
  - Occurs once every 30 minutes during Mode 1 (when BLE CH$\leftrightarrow$Edge communication is active)

- If the 30-minute mark occurs during Mode 2, the protocol waits until the next Mode 1 cycle
- Continues within the BLE communication slot until completion, potentially extending across multiple Mode 1 cycles
- Allows lost BLE devices to reintegrate with the network
- Re-executes full BLE discovery process including:
  * Cluster reformation to optimize network topology
  * BLE Clusterhead re-election in case better-connected nodes are available
  * Path re-establishment for improved routing
  * Neighbor identification updates
- Normal Mode 2 cycles continue during BLE Rediscovery (LoRA communication unaffected)

• **Submode 2 - LoRA Rediscovery**:
  - Occurs once every 30 minutes during Mode 2 (when LoRA CH↔Edge communication is active)
  - If the 30-minute mark occurs during Mode 1, the protocol waits until the next Mode 2 cycle
  - Continues within the LoRA communication slot until completion, potentially extending across multiple Mode 2 cycles
  - Allows lost LoRA devices to reintegrate with the network
  - Re-executes full LoRA discovery process including:
    * Channel re-assessment and potential migration
    * LoRA Clusterhead re-election for optimal connectivity
    * Crowding factor recalculation
    * Cluster reformation based on current network conditions
  - Normal Mode 1 cycles continue during LoRA Rediscovery (BLE communication unaffected)

The 30-minute rediscovery interval ensures that lost or new devices can rejoin the network and that network topology remains optimized without excessive overhead from continuous discovery operations. The maximum delay before rediscovery begins is 1.5 seconds (one mode cycle).

2.5 Local Failure Handling and Network Resilience

While full network rediscovery and topology optimization occurs every 30 minutes, the protocol includes local workarounds and redundancies to handle critical failures that occur between rediscovery cycles:

- **Immediate Failover Mechanisms**: When a Clusterhead becomes unresponsive, Edge nodes can:
  - Detect the failure within several communication cycles (approximately 3-6 seconds)
  - Temporarily route through alternative paths using their cached topology information
  - Connect to adjacent Clusterheads that have overlapping coverage
  - Queue messages locally until connectivity is restored
- **Redundant Path Maintenance**: During normal operation:
  - Edge nodes maintain awareness of multiple nearby Clusterheads (not just their primary)
  - Backup routing paths are cached from discovery message PSF (Path So Far) fields
  - Nodes track RSSI values from multiple Clusterheads for rapid failover decisions
- **Emergency Re-election**: In cases of complete Clusterhead failure:
  - Nearby high-connectivity nodes can assume temporary Clusterhead responsibilities
  - Emergency elections use simplified metrics (direct neighbor count only)
  - Temporary Clusterheads serve until the next 30-minute rediscovery validates or replaces them

- **Distinction Between Failure Recovery and Optimization**:
  - **Local failure handling** (seconds to minutes): Addresses immediate connectivity issues, node failures, and maintains basic network functionality
  - **Full rediscovery** (30-minute intervals): Performs comprehensive topology optimization, including detection of better-connected nodes that would serve as superior Clusterheads, network-wide load rebalancing, and integration of new high-capacity devices

This two-tier approach ensures network resilience for immediate failures while preventing the overhead of continuous full-scale rediscovery. Critical failures are handled locally within seconds, while major topology improvements (such as promoting a newly-joined well-connected node to Clusterhead) occur during scheduled 30-minute rediscovery cycles.

3 Discovery

The overall goal of a single discovery cycle in both Bluetooth and LoRA is to form dynamic clusters for all nodes that are in sync with the current cycle. On a smaller scale, the goal for any given LoRA node is for it to know around 90% of the nodes within its given cluster. Any given device will also keep count of how many other devices they are directly connected to as well as the number of devices currently in the mesh. For BLE specifically, a device will only keep count of paths to critical nodes.

3.1 Bluetooth Discovery

Bluetooth Discovery occurs on a single channel. All devices/nodes will propagate a discovery message with the following format:

- **ID**: The unique identifier of the message sender
- **TTL**: Time To Live - the number of hops remaining before the message is discontinued
- **PSF**: Path So Far - the sequence of node IDs the message has traveled through
- **LHGPS**: Last Heard GPS location of the message sender. Each device caches its most recent GPS location and includes it in all discovery messages. If a device has never obtained a GPS lock (e.g., indoors, GPS-denied environments), the LHGPS field is marked as unavailable, and GPS-based proximity filtering is skipped for that device's messages.

During Bluetooth Discovery, each discovery cycle contains four message slots. As a result, a given node will use one slot to send its own discovery message (if applicable) and has up to 3 remaining slots available to forward other nodes' messages. We use three separate metrics in order to determine which messages to forward if there are more messages queued than available slots.

1. Picky Forwarding: Uses the Crowding factor to forward only a given percentage of messages, preventing network congestion.
2. GPS Proximity: If the LHGPS of the previous sender is too close to the current location of the device, it will not forward.
3. Time to Live: The remaining messages are ordered using TTL and the top 3 messages are chosen.

These metrics are chosen in order to throttle the amount of messages being propagated within the network. They are chosen to work ideally with a single Bluetooth cluster having a maximum capacity of 150 devices.

3.2 LoRA Discovery

LoRA Discovery depends on the following variables:

- **k**: the current channel the device is communicating on
- **ID**: the unique device identifier
- **Home**: the device's home channel
- **Crowding**: the crowding factor of the device

The variables k and Home both start at 1 initially. The protocol supports up to 56 LoRA channels with a capacity of around 32 devices per channel.

During LoRA Discovery, every node will first listen to channel k. We use RSSI and CAD as metrics to determine the next course of action. There are three possible outcomes:

Outcome 1 - Channel Overload: If too many devices are trying to communicate simultaneously, CAD will detect activity but no decipherable messages can be received, and RSSI will be extremely high. When high RSSI is detected, the device will automatically jump to a higher channel k' and repeat this process.

Outcome 2 - Channel Near Capacity: If RSSI is in an acceptable range, CAD is used in conjunction with RSSI to calculate a Crowding factor. This crowding factor determines whether the current channel has reached capacity. If a node's Crowding factor is high enough, it sends out a vote proposing a channel migration and querying other nodes if they wish to move up. If enough nodes hear the vote and concur, they signal their agreement, and all nodes that voted yes move up together to a new channel.

Outcome 3 - Channel Below Capacity: If the device has an acceptable Crowding factor and does not hear a vote, it knows that locally the mesh on channel k is below capacity and remains on that channel.

4 Clusterhead Discovery and Election

The overall structure of our hybrid mesh is that all devices will be on a Bluetooth mesh, with BLE Clusterheads communicating among themselves through the BLE mesh as much as possible. For any BLE Clusterheads that cannot be reached through the BLE mesh, they will become a LoRA node and will communicate within the LoRA mesh network. As a result, it is crucial that we choose well-connected BLE Clusterheads in order to minimize the number of nodes in our LoRA mesh, which is subject to extremely slow data transmission speeds.

4.1 Clusterhead Discovery and Election for Bluetooth

BLE Clusterhead election occurs during normal Bluetooth Discovery and is based on connectivity metrics to ensure well-connected nodes become Clusterheads. The election process has 3 rounds to ensure redundancy strength and works as follows:

1. **Connectivity Metric Calculation**: During the discovery phase, each node tracks:
   - Number of direct neighbors (nodes within 1 hop)
   - Number of unique paths discovered through forwarded messages
   - Geographic distribution of neighbors based on LHGPS data
2. **Process Description**: The beginning of discovery operates in the following order:
   - Crowding Factor: During a specificied period, each device will "noisy broadcast" - send a broadcast base level noise message. Each individual device will pick a stochastically randomized time slot to listen. When listening, each device will determine crowding factor as a function f(RSSI).
   - : Clusterhead Broadcast: Nodes will then stochastically transmit a "broadcast" in a small timeslot. Each node will listen the majority of the time, documenting how many nodes heard. This counts as "direct connection" count.
   - Clusterhead Candidacy: Each bluetooth node will elect itself a clusterhead or edge, determined by a function of the ratio between direct connection count and noise heard. The motivation for using ratio lies on two principles:
     - 1) RSSI will be greater than the contribution from direct connections. RSSI will account for noise from near devices that don't have a strong enough signal to parse, but still contribute noise
     - 2) RSSI will factor in "collision signals" as well. When signals collide while broadcasting connection count, this lowers connection count while still amplifying RSSI. As a result, since fur-

ther connections will use the same hash function parameters, nodes with a lower likelihood of nearby neighbors colliding will have higher direct connection count : noise heard ratios, making them more suitable for clusterheads. As a result, the more reachable direct connections in a noisy area a node has, the more likely it is to become clusterhead.

- 3) Clusterhead Comparison & Election Announcement: Clusterheads will broadcast their direct device count to remove nearby lower count clusterheads.
- Cluster Formation: Edge devices will orient themselves with a specific clusterhead and store the path and knowledge of the clusterhead. Cluster edges will then, using stochastic slots, broadcast themselves to feasible clusters, as well as paths to other clusterheads they have heard.
- Affirmation: Clusterheads, upon hearing from cluster edges, send out a propogating message to all edges to acknowledge that they have been stored to the clusterhead.
- Clusterhead Exchange: Clusterheads communicate between themselves using known paths to exchange presence messages to each other
- Future Operations - Path Maintenance (maintenance, not part of initial discovery): Affirms will maintain these paths and update paths, known clusterheads, known edges, and edges that have fallen out of feasible reach.

3. **Clusterhead Candidacy**: A node becomes a Clusterhead candidate if:
   - It has more than a threshold number of direct connections (dynamically determined based on local crowding factor)
   - Its neighbors are geographically well-distributed (not all clustered in one area)
   - It has successfully received and forwarded messages during discovery
4. **Election Announcement**: Nodes that meet the candidacy criteria announce their Clusterhead status in subsequent discovery cycles by

adding a Clusterhead flag to their discovery messages. Election announcement will happen 3 times in 3 separate rounds to ensure high stochastic probability of reaching all devices.

Devices will broadcast their direct neighbor count to resolve conflicts between clusters. The election announcement message will be structured in the following manner:

- **Class ID**: The class ID of a clusterhead class to self-identify as a (potential) clusterhead
- **ID**: The unique identifier of the message sender
- **PDSF**: Predicted devices reached so far. This function is computed by doing the summation of devices reached at each step. General structure:

  $$t(x) = \sum_i \Pi_i(\mathrm{x}_i)$$

  where x represents the direct connection count at each step i, excluding the device(s) message has reached previously.
- **PSF**: Path So Far - the sequence of node IDs the message has traveled through
- **Score**: Clusterhead's score as determined by candidacy.
- **Hash h(ID)**: Unique FDMA / TDMA Hash Function for edges to reach that specific clusterhead.

Election announcements will "flood" areas by propogation outwards from clusterhead through retransmission. When PDSF reaches cluster capacity, devices will stop retransmitting the election announcement, and no more devices will add themselves to the clusterhead.

5. **Conflict Resolution**: If multiple Clusterhead candidates have overlapping coverage:
   - Nodes compare the number of direct connections
   - The candidate with more direct connections becomes the Clusterhead
   - In case of ties, the node with the lower device ID takes precedence

6. **Election Re-announcement** Nodes that initially qualified for candidacy but heard a clusterhead candidate that had a higher direct connection count will re-announce upon the follwing cycles that they have renounced their candidacy.

7. **Cluster Formation**: Once Clusterheads are established, Edge nodes align themselves with the Clusterhead from which they received the lower length path or the highest "direct count" message. Devices will "memorize" all available paths to the clusterheads, and construct a tree to the clusterhead using available paths. In the case of multiple interconnected paths to one cluster, Djikstra's will be used to compute the shortest path - this will be added to feasible paths as well the "memorized path," all of which will be used as feasible paths to the target clusterhead.
   Upon completion, edge devices will use FDMA/TDMA slotting based on stochastic initital hash to reach the clusterhead.
   The message will be structured as following:

   - Edge Class Identifier
   - Edge ID
   - Path Taken (backwards of path to the edge)
   - GPS
   - TTL

   Upon hearing this, clusterheads will repeat propogation method, with a more fine-tuned hash to give a more accurate hash to all edges, as well affirm that edges have been heard.
   The affirmation message will be structured as following

   - Clusterhead Class Identifier
   - Clusterhead Device ID
   - DFS Appended List of Edge IDs
   - TTL
   - Cluster Occupancy: The number of devices currently in the cluster.

   Regarding paths to other clusterheads: additionally, at this stage, devices that have been found by multiple "clusterheads" will use their determined paths to relay information about all the clusterheads they are connected to - this information will be sent to each individual clusterhead. Clusterheads will use a few shortest-hop paths to determine the viable paths to any other clusterhead.

8. **Clusterhead Info-Passing**: Clusterheads, upon finding out how to reach other clusterheads, will relay all data about themselves and any clusterheads they know of via a managed flood using the paths to other clusterheads. From this, Djikstra's is run again to determine shortest path to any individual clusterhead from any other clusterhead.
9. **Path Maintenance**: Clusterheads maintain a list of paths to most nodes within their cluster (targeting 90% coverage) and paths to other Clusterheads for inter-cluster communication.
10. **Mapping**: Edges reach clusterheads upon first discovery, and re-reach the clusterheads every rediscovery or affirmation cycle. Clusterheads, in the same way, will re-reach other clusterheads every affirmation cycle. Affirms do not replace discovery, but:
    - allow for new clusterheads to announce
    - allow reorientation to new or alternate clusterheads
    - update edge devices on known clusterhead paths
    - update clusterheads on known edge devices
    - update clusterheads of each others' presence
11. **Result**: Each edge node will know: the path to its clusterhead, its communication hash, and the direct nearest neighbors for future use. Each clusterhead will have a table of edge nodes : paths in its respective cluster, and clusterhead: paths for the entire interconnected mesh.

4.2 Clusterhead Discovery and Election for LoRA

LoRA Clusterhead election occurs during LoRA Discovery and prioritizes well-connected nodes to minimize inter-cluster communication overhead. The election process works as follows:

1. **Connectivity Tracking**: During LoRA Discovery, each LoRA node monitors:
    - Number of direct LoRA neighbors within its channel
    - Signal strength (RSSI) from each neighbor
    - Successful message exchanges during discovery
2. **Clusterhead Candidacy**: A LoRA node becomes a Clusterhead candidate if:
    - It has a number of direct connections exceeding the local crowding factor threshold
    - It has stable connectivity (consistent RSSI values across multiple cycles)
    - It successfully participates in channel voting processes
3. **Election Announcement**: LoRA nodes that meet candidacy criteria announce their Clusterhead status by broadcasting a Clusterhead Declaration message containing their connection count and node ID.
4. **Conflict Resolution**: If a LoRA node receives multiple conflicting Clusterhead messages:
    - It compares the connection counts from each candidate
    - Joins the cluster of the Clusterhead with more direct connections
    - In case of ties, the node with the lower device ID takes precedence
5. **Cluster Formation**: Once Clusterheads are established, LoRA Edge nodes associate with their chosen Clusterhead and maintain this association across discovery cycles.
6. **Channel Overflow Management**: If there are too many LoRA nodes within a small area exceeding channel capacity, nodes will initiate a

voting process to create additional channels to accommodate the increased density.

5 Within Network Searching

5.1 Searching for a Node

The search will propagate upwards through the network hierarchy. When a BLE Edge wishes to find another node, the search proceeds as follows:

1. The BLE Edge queries its BLE Clusterhead
2. The BLE Clusterhead checks if the target node is within its own cluster (it knows most devices in its cluster):
   - If yes: returns affirmation with path to the target
   - If no: proceeds to step 3
3. The BLE Clusterhead queries other BLE Clusterheads within the same LoRA cluster (i.e., BLE Clusterheads connected to the same LoRA node)
4. If still not found, the query is escalated to the LoRA node, which queries other LoRA nodes within its LoRA cluster
5. If the target is still not found, the Search Request propagates to the LoRA Clusterhead level for inter-cluster searching

These will be Search Requests that travel between LoRA Clusterheads. See the later section on LoRA Clusterhead to LoRA Clusterhead communication for more detail on Search Requests.

5.2 Node is Unreachable

If a given device is unreachable, which means that after two cycles of searching, the desired device is still not found, the message will propagate back down until it reaches the node that requested the search. This node will then wait for three additional cycles before repeating. If the device is still not found at this point, then the searching device will display a message stating that the desired device is outside the boundaries of the current mesh network.

6 Within Network Communication

Once the network topology is established through discovery and clusterhead election, devices transition to Communication Mode for data transmission. Communication occurs through time-slotted messaging coordinated by hash functions distributed by Clusterheads, ensuring collision-free transmission and efficient bandwidth utilization across the hybrid mesh.

6.1 Hash Function Distribution and Time Slot Assignment

During discovery, a BLE Clusterhead will calculate its own hash function and provide that to each BLE Edge within its cluster. When a BLE Edge receives the hash function, it will calculate its time slots using its device ID and the hash function. These time slots are specifically meant for listening: that is any incoming messages to the BLE Edge will occur during this time slot. As a result, within a given BLE messaging phase, there will be three iterations of the time slotted messaging in order to ensure that messages reach the desired Edge or Clusterhead probabilistically.

6.2 BLE Edge to BLE Edge Communication

This is a very rare case and only occurs in two instances: either a BLE Edge is trying to communicate with another BLE Edge within its own mesh, or this communication is serving as a bridge for a BLE Clusterhead to BLE Edge/Clusterhead message as part of the BLE mesh. In the first case, the BLE Edge will just locally flood its own cluster in order to communicate. In the second case, the BLE Edge will simply add the message to the queue of messages it needs to forward and may either be forwarded or discarded accordingly.

6.3 BLE Edge to BLE Clusterhead Communication

Since a BLE Edge will most likely have received an Affirmation from the BLE Clusterhead, which also tracks the path it took to get from the Clusterhead to the BLE Edge, the BLE Edge can simply reverse that path to return back to the BLE Clusterhead. During a given Messaging Phase, the BLE Clusterhead will be listening for around 75% of the time and messaging for the remaining 25%. As a result, all BLE Edges that wish to reach the BLE Clusterhead will send messages through the BLE mesh with a given path to the BLE Clusterhead. The messages will propagate through the nodes on the path until it reaches the BLE Clusterhead.

6.4 BLE Clusterhead to BLE Edge Communication

In a given BLE cluster, a single BLE Clusterhead will not be directly connected to all the BLE Edges within the cluster. As a result, the BLE Clusterhead will instead send a single long message addressed to every node in the cluster. The message content will consist of multiple chains of BLE Edge IDs and the message meant for that device, with a blank placeholder for no message. When a BLE Edge receives this message, it will then trim the message after its own message before propagating the trimmed message.

The BLE Clusterhead will also send a hash function in the message as well. This hash function will take in the device ID and return a time slot, which will tell a device when to propagate its message in order to prevent overlap. This is so that messages can propagate through multiple paths without overlap to ensure they reach all intended recipients.

6.5 LoRA Augmentation

LoRA augmentation provides a fallback and extension mechanism for BLE communication when devices are beyond Bluetooth range or when the BLE mesh becomes congested. The augmentation strategy works as follows:

- **Automatic Fallback**: When a BLE Clusterhead cannot reach another BLE Clusterhead via the Bluetooth mesh, it automatically transitions to become a LoRA node to maintain network connectivity.
- **Hybrid Operation**: Devices can simultaneously maintain both BLE and LoRA connectivity, operating as BLE Edges within their local cluster while their BLE Clusterhead communicates via LoRA with distant clusters.
- **Message Bridging**: LoRA nodes receive aggregated messages from their BLE Clusterheads and forward them through the LoRA mesh to distant LoRA Clusterheads, which then distribute the messages to their respective BLE clusters.
- **Bandwidth Management**: Due to LoRA's lower data rates, messages are prioritized by urgency and compressed when possible. High-frequency location updates may be throttled when transmitted over LoRA.
- **Network Expansion**: LoRA extends the effective range of the overall mesh network significantly beyond what BLE alone could achieve, enabling communication across larger geographical areas while maintaining the high-density, low-latency benefits of BLE for local communication.

6.6 LoRA Edge to LoRA Clusterhead Communication

This uses a similar strategy to the BLE communication in that every device in a LoRA cluster has a slot for receiving messages. This slot is, instead, determined stochastically based on the number of devices in the cluster.

6.7 LoRA Clusterhead to LoRA Edge Communication

Since each LoRA Clusterhead is a direct neighbor with all the LoRA Edges within its cluster, when it is time to message LoRA Edges, the Clusterhead will send a single long message. All LoRA Edges will receive this message and simply extract the part of the message that is addressed to them.

6.8 LoRA Clusterhead to LoRA Clusterhead Communication

There are two types of messages that will be passed between LoRA Clusterheads: Search Requests, which will be addressed to all LoRA Clusterheads; and 1-to-1 Messages, which are communications from a single LoRA Clusterhead to another.

Search Requests are when a LoRA Clusterhead needs to find a device that is not within the hybrid mesh of its entire cluster. The LoRA Clusterhead will send a Search Request, which will flood the LoRA Clusterhead mesh. We use similar principles to our BLE Discovery, with managed flooding based on the crowding as well as time to live on search requests.

The Search Request message format is:

- **Target ID**: The unique identifier of the node being searched for
- **Source ID**: The unique identifier of the node that initiated the search
- **Time to Live**: The number of hops remaining before the search request expires
- **Path So Far**: The sequence of node IDs the search request has traveled through

Every LoRA node should have a list of BLE nodes within their corresponding BLE cluster as well as the paths to those nodes. Every LoRA Clusterhead will also have a list of all other LoRA Clusterheads in the mesh as well as paths to each of those Clusterheads.

As a result, when a LoRA Clusterhead receives a search request where they are not part of the source, it will then query all LoRA nodes within its cluster. The LoRA nodes will then query all of their corresponding BLE Clusterheads, who have a list of most nodes within their cluster as well as a path to that BLE node. If a match is found, the BLE Clusterhead informs the LoRA Edge of the BLE node's presence as well as the path to that BLE node, which in turn informs its LoRA Clusterhead. The LoRA Clusterhead then knows the path from the BLE node to itself as well as the path from itself to the source LoRA Clusterhead. The source LoRA Clusterhead is then informed about this path, which it then relays to the original node that requested the search.

In the case of 1-to-1 messages, these occur when a path to the target device is already known. The LoRA Clusterhead will use a hash function to assign different LoRA Edges within its cluster different time slots. These time slots will be used for the LoRA Clusterhead to send any messages meant for a given device to the device. The LoRA Clusterheads themselves will also be time slotted. This means that any 1-to-1 messages to be delivered to any BLE or LoRA node in another mesh will be gathered and sent as a single long message to the relevant LoRA Clusterhead during their time slot.

7 Rediscovery

Rediscovery can occur due to a multitude of issues. A device may be powered off, it may leave its existing cluster, or it may be a completely new device joining an existing mesh. Regardless, all of these "Lost Devices" will undergo the same process in Bluetooth and LoRA to reintegrate with the existing mesh.

7.1 Bluetooth Rediscovery

In Bluetooth, all devices will send a ping at the beginning of the Bluetooth Discovery Phase. This ping message will include a message ID, a device ID, and the GPS location. Any Lost Devices that hear this ping message will use this ping to determine their nearest neighbor using the GPS coordinates. Then, they will piggyback off of the slots of their nearest neighbor for a single iteration before being fully incorporated into the mesh.

7.2 LoRA Rediscovery

In LoRA, a device will begin decrementing its Home channel when it does not hear any other devices on it. As a result, most Lost Devices will have reverted to being on the lowest home channel when they are able to connect to an existing mesh. When a Lost Device encounters a LoRA mesh, it will first listen to identify k, the number of total active LoRA channels. Once it has done so, it will join one of those k channels. If there are too many devices on a given channel, it will raise its home channel. If this fails, then it will transmit a vote to raise the k total channels by one. From here on, a similar process follows to LoRA Discovery.

Once a Lost Device is settled in a home channel, it will message the Clusterhead to inform the Clusterhead that it is now a part of its cluster. If the Clusterhead hears the message, it will send a message at the end of the LoRA Discovery Phase affirming the members of its cluster. The Lost Device will retransmit to the Clusterhead three times, once per cycle, and wait for an affirmation message. If it does not receive one after three cycles, then it will attempt to join another cluster using this same process.

8 Timings and Calculations